

Code output

```

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[Icons]
al>
Water-Allocation 1.0.c
1 #include <stdio.h>
2 #include <string.h>
3
4 #define MAX_ZONES
5 #define MAX_NAME
6 #define EPSILON
7
8 typedef struct {
9     char zoneID[10];
10    char serviceZoneName[50];
11    char type[25];
12    double minReq;
13    double lossPer;
14    int prevWaitingCycles;
15
16    double criticality;
17    double priority;
18    int priorityOrder;
19
20    double allocated;
21    double effective;
22    double shortage;
23    char condition[20];
24
25 } Zone;

```

```

25.03_ case1_input.txt all_test_outputs.txt
File Edit View H1 B I S [Icons]
##### CASE 7 FULL OUTPUT #####
=== Emergency Potable Water Allocation System ===

Enter total available potable water at the distribution centre (litres): Enter number of service zones (max 50):
--- Zone 1 ---
Zone ID: Service zone name: Type (Hospital/EmergencyShelter/Residential/EmergencyService): Requested water (L): Minimum requirement (L): Distribution loss (%):
Previous waiting cycles:
--- Zone 2 ---
Zone ID: Service zone name: Type (Hospital/EmergencyShelter/Residential/EmergencyService): Requested water (L): Minimum requirement (L): Distribution loss (%):
Previous waiting cycles:
--- Zone 3 ---
Zone ID: Service zone name: Type (Hospital/EmergencyShelter/Residential/EmergencyService): Requested water (L): Minimum requirement (L): Distribution loss (%):
Previous waiting cycles:
--- Zone 4 ---
Zone ID: Service zone name: Type (Hospital/EmergencyShelter/Residential/EmergencyService): Requested water (L): Minimum requirement (L): Distribution loss (%):
Previous waiting cycles:
--- Zone 5 ---
Zone ID: Service zone name: Type (Hospital/EmergencyShelter/Residential/EmergencyService): Requested water (L): Minimum requirement (L): Distribution loss (%):
Previous waiting cycles:
--- Zone 6 ---
Zone ID: Service zone name: Type (Hospital/EmergencyShelter/Residential/EmergencyService): Requested water (L): Minimum requirement (L): Distribution loss (%):
Previous waiting cycles:
--- Zones ordered by allocation priority ---
Rank Service Zone Requested MinReq Alloc EffDelv Shortage Condition
-----
1 Municipal Hospital 4000.0 3000.0 3061.2 3000.0 1000.0 MinimumSatisfied
2 Ward 5 Residential Area 3800.0 1800.0 2045.5 1800.0 2000.0 MinimumSatisfied
3 Central Flood Shelter 3500.0 2500.0 2604.2 2500.0 1000.0 MinimumSatisfied
4 School Emergency Shelter 2600.0 1600.0 1684.2 1600.0 1000.0 MinimumSatisfied
5 Fire and Emergency Service 2000.0 1500.0 1515.2 1500.0 500.0 MinimumSatisfied
6 Ward 3 Residential Area 4500.0 2000.0 1489.8 1370.6 3129.4 BelowMinimum

===== SUMMARY OF ALLOCATION =====
Zone ID | Service Zone | Requested | MinReq | Priority | Allocated | EffDelivered | Shortage | Condition
-----|-----|-----|-----|-----|-----|-----|-----|-----
Z01 | Municipal Hospital | 4000.0 | 3000.0 | 1 | 3061.2 | 3000.0 | 1000.0 | MinimumSatisfied
Z04 | Ward 5 Residential Area | 3800.0 | 1800.0 | 2 | 2045.5 | 1800.0 | 2000.0 | MinimumSatisfied
Z02 | Central Flood Shelter | 3500.0 | 2500.0 | 3 | 2604.2 | 2500.0 | 1000.0 | MinimumSatisfied
Z05 | School Emergency Shelter | 2600.0 | 1600.0 | 4 | 1684.2 | 1600.0 | 1000.0 | MinimumSatisfied
Z06 | Fire and Emergency Service | 2000.0 | 1500.0 | 5 | 1515.2 | 1500.0 | 500.0 | MinimumSatisfied
Z03 | Ward 3 Residential Area | 4500.0 | 2000.0 | 6 | 1489.8 | 1370.6 | 3129.4 | BelowMinimum

Total water available at start : 12400.0 L
Total water requested : 20400.0 L
Total water allocated (released) : 12400.0 L
Total effective water delivered : 11770.6 L

```

```
Water-Allocation 1.0.c
1 #include <stdio.h>
2 #include <string.h>
3
4 #define MAX_ZONES
5 #define MAX_NAME
6 #define EPSILON
7
8
9 typedef struct {
10     char zoneID[10];
11     char serviceZone[50];
12     char type[25];
13     double minReq;
14     double lossPer;
15     int prevWait;
16
17     double critical;
18     double priority;
19     int priority;
20
21     double allocated;
22     double effective;
23     double shortage;
24     char condition;
25 } Zone;
```

```
case1_input.txt all_test_outputs.txt
File Edit View H1 B I S G A
##### CASE 6 FULL OUTPUT #####
=== Emergency Potable Water Allocation System ===

Enter total available potable water at the distribution centre (litres): Enter number of service zones (max 50):
--- Zone 1 ---
Zone ID: Service zone name: Type (Hospital/EmergencyShelter/Residential/EmergencyService): Requested water (L): Minimum requirement (L): Distribution loss (%):
Previous waiting cycles:
--- Zone 2 ---
Zone ID: Service zone name: Type (Hospital/EmergencyShelter/Residential/EmergencyService): Requested water (L): Minimum requirement (L): Distribution loss (%):
Previous waiting cycles:
--- Zone 3 ---
Zone ID: Service zone name: Type (Hospital/EmergencyShelter/Residential/EmergencyService): Requested water (L): Minimum requirement (L): Distribution loss (%):
Previous waiting cycles:
--- Zone 4 ---
Zone ID: Service zone name: Type (Hospital/EmergencyShelter/Residential/EmergencyService): Requested water (L): Minimum requirement (L): Distribution loss (%):
Previous waiting cycles:
--- Zone 5 ---
Zone ID: Service zone name: Type (Hospital/EmergencyShelter/Residential/EmergencyService): Requested water (L): Minimum requirement (L): Distribution loss (%):
Previous waiting cycles:
--- Zone 6 ---
Zone ID: Service zone name: Type (Hospital/EmergencyShelter/Residential/EmergencyService): Requested water (L): Minimum requirement (L): Distribution loss (%):
Previous waiting cycles:
--- Zones ordered by allocation priority ---
Rank Service Zone Requestd MinReq Alloc EffDlv Shortage Condition
-----
1 Ward 3 Residential Area 4500.0 2000.0 2589.8 2382.6 2117.4 MinimumSatisfied
2 Municipal Hospital 4000.0 3000.0 3061.2 3000.0 1000.0 MinimumSatisfied
3 Ward 5 Residential Area 3800.0 1800.0 2045.5 1800.0 2000.0 MinimumSatisfied
4 Central Flood Shelter 3500.0 2500.0 2604.2 2500.0 1000.0 MinimumSatisfied
5 School Emergency Shelter 2600.0 1600.0 1684.2 1600.0 1000.0 MinimumSatisfied
6 Fire and Emergency Service 2000.0 1500.0 1515.2 1500.0 500.0 MinimumSatisfied

===== SUMMARY OF ALLOCATION =====
Zone ID | Service Zone | Requested | MinReq | Priority | Allocated | EffDelivered | Shortage | Condition
-----
Z03 | Ward 3 Residential Area | 4500.0 | 2000.0 | 1 | 2589.8 | 2382.6 | 2117.4 | MinimumSatisfied
Z01 | Municipal Hospital | 4000.0 | 3000.0 | 2 | 3061.2 | 3000.0 | 1000.0 | MinimumSatisfied
Z04 | Ward 5 Residential Area | 3800.0 | 1800.0 | 3 | 2045.5 | 1800.0 | 2000.0 | MinimumSatisfied
Z02 | Central Flood Shelter | 3500.0 | 2500.0 | 4 | 2604.2 | 2500.0 | 1000.0 | MinimumSatisfied
Z05 | School Emergency Shelter | 2600.0 | 1600.0 | 5 | 1684.2 | 1600.0 | 1000.0 | MinimumSatisfied
Z06 | Fire and Emergency Service | 2000.0 | 1500.0 | 6 | 1515.2 | 1500.0 | 500.0 | MinimumSatisfied

Total water available at start : 13500.0 L
Total water requested : 20400.0 L
Total water allocated (released) : 13500.0 L
Total effective water delivered : 12782.6 L
```

```
Water-Allocation 1.0.c
1 #include <stdio.h>
2 #include <string.h>
3
4 #define MAX_ZONES
5 #define MAX_NAME
6 #define EPSILON
7
8
9 typedef struct {
10     char zoneID[10];
11     char serviceZone[50];
12     char type[25];
13     double minReq;
14     double lossPer;
15     int prevWait;
16
17     double critical;
18     double priority;
19     int priority;
20
21     double allocated;
22     double effective;
23     double shortage;
24     char condition;
25 } Zone;
```

```
case1_input.txt all_test_outputs.txt
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##### CASE 5 FULL OUTPUT #####
=== Emergency Potable Water Allocation System ===

Enter total available potable water at the distribution centre (litres): Enter number of service zones (max 50):
--- Zone 1 ---
Zone ID: Service zone name: Type (Hospital/EmergencyShelter/Residential/EmergencyService): Requested water (L): Minimum requirement (L): Distribution loss (%):
Previous waiting cycles:
--- Zone 2 ---
Zone ID: Service zone name: Type (Hospital/EmergencyShelter/Residential/EmergencyService): Requested water (L): Minimum requirement (L): Distribution loss (%):
Previous waiting cycles:
--- Zone 3 ---
Zone ID: Service zone name: Type (Hospital/EmergencyShelter/Residential/EmergencyService): Requested water (L): Minimum requirement (L): Distribution loss (%):
Previous waiting cycles:
--- Zone 4 ---
Zone ID: Service zone name: Type (Hospital/EmergencyShelter/Residential/EmergencyService): Requested water (L): Minimum requirement (L): Distribution loss (%):
Previous waiting cycles:
--- Zone 5 ---
Zone ID: Service zone name: Type (Hospital/EmergencyShelter/Residential/EmergencyService): Requested water (L): Minimum requirement (L): Distribution loss (%):
Previous waiting cycles:
--- Zone 6 ---
Zone ID: Service zone name: Type (Hospital/EmergencyShelter/Residential/EmergencyService): Requested water (L): Minimum requirement (L): Distribution loss (%):
Previous waiting cycles:
--- Zones ordered by allocation priority ---
Rank Service Zone Requestd MinReq Alloc EffDlv Shortage Condition
-----
1 Municipal Hospital 4000.0 3000.0 3061.2 3000.0 1000.0 MinimumSatisfied
2 Ward 5 Residential Area 3800.0 1800.0 2045.5 1800.0 2000.0 MinimumSatisfied
3 Central Flood Shelter 3500.0 2500.0 2604.2 2500.0 1000.0 MinimumSatisfied
4 School Emergency Shelter 2600.0 1600.0 1684.2 1600.0 1000.0 MinimumSatisfied
5 Fire and Emergency Service 2000.0 1500.0 1515.2 1500.0 500.0 MinimumSatisfied
6 Ward 3 Residential Area 4500.0 2000.0 135.2 124.4 4375.6 BelowMinimum

===== SUMMARY OF ALLOCATION =====
Zone ID | Service Zone | Requested | MinReq | Priority | Allocated | EffDelivered | Shortage | Condition
-----
Z01 | Municipal Hospital | 4000.0 | 3000.0 | 1 | 3061.2 | 3000.0 | 1000.0 | MinimumSatisfied
Z04 | Ward 5 Residential Area | 3800.0 | 1800.0 | 2 | 2045.5 | 1800.0 | 2000.0 | MinimumSatisfied
Z02 | Central Flood Shelter | 3500.0 | 2500.0 | 3 | 2604.2 | 2500.0 | 1000.0 | MinimumSatisfied
Z05 | School Emergency Shelter | 2600.0 | 1600.0 | 4 | 1684.2 | 1600.0 | 1000.0 | MinimumSatisfied
Z06 | Fire and Emergency Service | 2000.0 | 1500.0 | 5 | 1515.2 | 1500.0 | 500.0 | MinimumSatisfied
Z03 | Ward 3 Residential Area | 4500.0 | 2000.0 | 6 | 135.2 | 124.4 | 4375.6 | BelowMinimum

Total water available at start : 13500.0 L
Total water requested : 20400.0 L
Total water allocated (released) : 13500.0 L
Total effective water delivered : 10524.4 L
Total unresolved requirement : 9875.6 L
Remaining water at distribution centre: 0.0 L
Zones fully satisfied : 0
Zones with only minimum satisfied : 5
Zones below minimum requirement : 1
```


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25.03_ case1_input.txt all_test_outputs.txt

CASE 4 FULL OUTPUT #####
=== Emergency Potable Water Allocation System ===

Enter total available potable water at the distribution centre (litres): Enter number of service zones (max 50):
--- Zone 1 ---
Zone ID: Service zone name: Type (Hospital/EmergencyShelter/Residential/EmergencyService): Requested water (L): Minimum requirement (L): Distribution loss (%):
Previous waiting cycles:
--- Zone 2 ---
Zone ID: Service zone name: Type (Hospital/EmergencyShelter/Residential/EmergencyService): Requested water (L): Minimum requirement (L): Distribution loss (%):
Previous waiting cycles:
--- Zone 3 ---
Zone ID: Service zone name: Type (Hospital/EmergencyShelter/Residential/EmergencyService): Requested water (L): Minimum requirement (L): Distribution loss (%):
Previous waiting cycles:
--- Zone 4 ---
Zone ID: Service zone name: Type (Hospital/EmergencyShelter/Residential/EmergencyService): Requested water (L): Minimum requirement (L): Distribution loss (%):
Previous waiting cycles:
--- Zones ordered by allocation priority ---

Rank	Service Zone	Reqstd	MinReq	Alloc	EffDlv	Shortage	Condition
1	Shelter Alpha	3000.0	2000.0	3157.9	3000.0	0.0	FullRequirement
2	Shelter Beta	3000.0	2000.0	3157.9	3000.0	0.0	FullRequirement
3	Municipal Hospital	4000.0	3000.0	4081.6	4000.0	0.0	FullRequirement
4	Fire and Emergency Service	2000.0	1500.0	2020.2	2000.0	0.0	FullRequirement

===== SUMMARY OF ALLOCATION =====

Zone ID	Service Zone	Requested	MinReq	Priority	Allocated	EffDelivered	Shortage	Condition
ZA	Shelter Alpha	3000.0	2000.0	1	3157.9	3000.0	0.0	FullRequirement
ZB	Shelter Beta	3000.0	2000.0	2	3157.9	3000.0	0.0	FullRequirement
Z01	Municipal Hospital	4000.0	3000.0	3	4081.6	4000.0	0.0	FullRequirement
Z06	Fire and Emergency Service	2000.0	1500.0	4	2020.2	2000.0	0.0	FullRequirement

Total water available at start : 13500.0 L
Total water requested : 12000.0 L
Total water allocated (released) : 12417.6 L
Total effective water delivered : 12000.0 L
Total unresolved requirement : 0.0 L
Remaining water at distribution centre: 1082.4 L
Zones fully satisfied : 4
Zones with only minimum satisfied : 0
Zones below minimum requirement : 0
Zones completely unserved : 0

Check: total allocated (12417.6) does not exceed available supply (13500.0). OK.

Zone with the highest unresolved requirement: ZA (Shelter Alpha) - shortage 0.0 L

Search for a zone by Zone ID? (1 = Yes, 0 = No): -e

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25.03_ case1_input.txt all_test_outputs.txt

CASE 3 FULL OUTPUT #####
=== Emergency Potable Water Allocation System ===

Enter total available potable water at the distribution centre (litres): Enter number of service zones (max 50):
--- Zone 1 ---
Zone ID: Service zone name: Type (Hospital/EmergencyShelter/Residential/EmergencyService): Requested water (L): Minimum requirement (L): Distribution loss (%):
Previous waiting cycles:
--- Zone 2 ---
Zone ID: Service zone name: Type (Hospital/EmergencyShelter/Residential/EmergencyService): Requested water (L): Minimum requirement (L): Distribution loss (%):
Previous waiting cycles:
--- Zone 3 ---
Zone ID: Service zone name: Type (Hospital/EmergencyShelter/Residential/EmergencyService): Requested water (L): Minimum requirement (L): Distribution loss (%):
Previous waiting cycles:
--- Zone 4 ---
Zone ID: Service zone name: Type (Hospital/EmergencyShelter/Residential/EmergencyService): Requested water (L): Minimum requirement (L): Distribution loss (%):
Previous waiting cycles:
--- Zone 5 ---
Zone ID: Service zone name: Type (Hospital/EmergencyShelter/Residential/EmergencyService): Requested water (L): Minimum requirement (L): Distribution loss (%):
Previous waiting cycles:
--- Zone 6 ---
Zone ID: Service zone name: Type (Hospital/EmergencyShelter/Residential/EmergencyService): Requested water (L): Minimum requirement (L): Distribution loss (%):
Previous waiting cycles:
--- Zones ordered by allocation priority ---

Rank	Service Zone	Reqstd	MinReq	Alloc	EffDlv	Shortage	Condition
1	Municipal Hospital	4000.0	3000.0	3061.2	3000.0	1000.0	MinimumSatisfied
2	Ward 5 Residential Area	3800.0	1800.0	2045.5	1800.0	2000.0	MinimumSatisfied
3	Central Flood Shelter	3500.0	2500.0	2604.2	2500.0	1000.0	MinimumSatisfied
4	School Emergency Shelter	2600.0	1600.0	1684.2	1600.0	1000.0	MinimumSatisfied
5	Fire and Emergency Service	2000.0	1500.0	604.9	598.9	1401.1	BelowMinimum
6	Ward 3 Residential Area	4500.0	2000.0	0.0	0.0	4500.0	Unserved

===== SUMMARY OF ALLOCATION =====

Zone ID	Service Zone	Requested	MinReq	Priority	Allocated	EffDelivered	Shortage	Condition
Z01	Municipal Hospital	4000.0	3000.0	1	3061.2	3000.0	1000.0	MinimumSatisfied
Z04	Ward 5 Residential Area	3800.0	1800.0	2	2045.5	1800.0	2000.0	MinimumSatisfied
Z02	Central Flood Shelter	3500.0	2500.0	3	2604.2	2500.0	1000.0	MinimumSatisfied
Z05	School Emergency Shelter	2600.0	1600.0	4	1684.2	1600.0	1000.0	MinimumSatisfied
Z06	Fire and Emergency Service	2000.0	1500.0	5	604.9	598.9	1401.1	BelowMinimum
Z03	Ward 3 Residential Area	4500.0	2000.0	6	0.0	0.0	4500.0	Unserved

Total water available at start : 10000.0 L
Total water requested : 20400.0 L
Total water allocated (released) : 10000.0 L
Total effective water delivered : 9498.9 L
Total unresolved requirement : 10901.1 L
Remaining water at distribution centre: 0.0 L
Zones fully satisfied : 0
Zones with only minimum satisfied : 4
Zones below minimum requirement : 1
Zones completely unserved : 1

Ln 73, Col 95 26,095 characters Plain text 70% Unix (LF) UTF-8


```
1 #include <stdio.h>
2 #include <string.h>
3
4 #define MAX_ZONES
5 #define MAX_NAME
6 #define EPSILON
7
8
9 typedef struct {
10     char zoneID[10];
11     char serviceZoneName[50];
12     char type[25];
13     double minRequirement;
14     double lossPercentage;
15     int previousWaitingCycles;
16
17     double criticalRequirement;
18     double priority;
19     int priorityOrder;
20
21     double allocatedWater;
22     double effectiveWater;
23     double shortage;
24     char condition[20];
25 } Zone;
```

```
25.03_ case1_input.txt all_test_outputs.txt
File Edit View H1 B I S G A
##### CASE 2 FULL OUTPUT #####
=== Emergency Potable Water Allocation System ===

Enter total available potable water at the distribution centre (litres): Enter number of service zones (max 50):
--- Zone 1 ---
Zone ID: Service zone name: Type (Hospital/EmergencyShelter/Residential/EmergencyService): Requested water (L): Minimum requirement (L): Distribution loss (%):
Previous waiting cycles:
--- Zone 2 ---
Zone ID: Service zone name: Type (Hospital/EmergencyShelter/Residential/EmergencyService): Requested water (L): Minimum requirement (L): Distribution loss (%):
Previous waiting cycles:
--- Zone 3 ---
Zone ID: Service zone name: Type (Hospital/EmergencyShelter/Residential/EmergencyService): Requested water (L): Minimum requirement (L): Distribution loss (%):
Previous waiting cycles:
--- Zone 4 ---
Zone ID: Service zone name: Type (Hospital/EmergencyShelter/Residential/EmergencyService): Requested water (L): Minimum requirement (L): Distribution loss (%):
Previous waiting cycles:
--- Zone 5 ---
Zone ID: Service zone name: Type (Hospital/EmergencyShelter/Residential/EmergencyService): Requested water (L): Minimum requirement (L): Distribution loss (%):
Previous waiting cycles:
--- Zone 6 ---
Zone ID: Service zone name: Type (Hospital/EmergencyShelter/Residential/EmergencyService): Requested water (L): Minimum requirement (L): Distribution loss (%):
Previous waiting cycles:
--- Zones ordered by allocation priority ---
Rank Service Zone Reqstd MinReq Alloc EffDlv Shortage Condition
-----
1 Municipal Hospital 4000.0 3000.0 4081.6 4000.0 0.0 FullRequirement
2 Ward 5 Residential Area 3800.0 1800.0 4318.2 3800.0 0.0 FullRequirement
3 Central Flood Shelter 3500.0 2500.0 3645.8 3500.0 0.0 FullRequirement
4 School Emergency Shelter 2600.0 1600.0 2736.8 2600.0 0.0 FullRequirement
5 Fire and Emergency Service 2000.0 1500.0 2020.2 2000.0 0.0 FullRequirement
6 Ward 3 Residential Area 4500.0 2000.0 4891.3 4500.0 0.0 FullRequirement

===== SUMMARY OF ALLOCATION =====
Zone ID | Service Zone | Requested | MinReq | Priority | Allocated | EffDelivered | Shortage | Condition
-----
Z01 | Municipal Hospital | 4000.0 | 3000.0 | 1 | 4081.6 | 4000.0 | 0.0 | FullRequirement
Z04 | Ward 5 Residential Area | 3800.0 | 1800.0 | 2 | 4318.2 | 3800.0 | 0.0 | FullRequirement
Z02 | Central Flood Shelter | 3500.0 | 2500.0 | 3 | 3645.8 | 3500.0 | 0.0 | FullRequirement
Z05 | School Emergency Shelter | 2600.0 | 1600.0 | 4 | 2736.8 | 2600.0 | 0.0 | FullRequirement
Z06 | Fire and Emergency Service | 2000.0 | 1500.0 | 5 | 2020.2 | 2000.0 | 0.0 | FullRequirement
Z03 | Ward 3 Residential Area | 4500.0 | 2000.0 | 6 | 4891.3 | 4500.0 | 0.0 | FullRequirement

Total water available at start : 22000.0 L
Total water requested : 20400.0 L
Total water allocated (released) : 21694.0 L
Total effective water delivered : 20400.0 L
Total unresolved requirement : 0.0 L
Remaining water at distribution centre: 306.0 L
Zones fully satisfied : 6
Zones with only minimum satisfied : 0
Zones below minimum requirement : 0
Zones completely unserved : 0
```

```
1 #include <stdio.h>
2 #include <string.h>
3
4 #define MAX_ZONES
5 #define MAX_NAME
6 #define EPSILON
7
8
9 typedef struct {
10     char zoneID[10];
11     char serviceZoneName[50];
12     char type[25];
13     double minRequirement;
14     double lossPercentage;
15     int previousWaitingCycles;
16
17     double criticalRequirement;
18     double priority;
19     int priorityOrder;
20
21     double allocatedWater;
22     double effectiveWater;
23     double shortage;
24     char condition[20];
25 } Zone;
```

```
25.03_ case1_input.txt all_test_outputs.txt
File Edit View H1 B I S G A
##### CASE 1 FULL OUTPUT #####
=== Emergency Potable Water Allocation System ===

Enter total available potable water at the distribution centre (litres): Enter number of service zones (max 50):
--- Zone 1 ---
Zone ID: Service zone name: Type (Hospital/EmergencyShelter/Residential/EmergencyService): Requested water (L): Minimum requirement (L): Distribution loss (%):
Previous waiting cycles:
--- Zone 2 ---
Zone ID: Service zone name: Type (Hospital/EmergencyShelter/Residential/EmergencyService): Requested water (L): Minimum requirement (L): Distribution loss (%):
Previous waiting cycles:
--- Zone 3 ---
Zone ID: Service zone name: Type (Hospital/EmergencyShelter/Residential/EmergencyService): Requested water (L): Minimum requirement (L): Distribution loss (%):
Previous waiting cycles:
--- Zone 4 ---
Zone ID: Service zone name: Type (Hospital/EmergencyShelter/Residential/EmergencyService): Requested water (L): Minimum requirement (L): Distribution loss (%):
Previous waiting cycles:
--- Zone 5 ---
Zone ID: Service zone name: Type (Hospital/EmergencyShelter/Residential/EmergencyService): Requested water (L): Minimum requirement (L): Distribution loss (%):
Previous waiting cycles:
--- Zone 6 ---
Zone ID: Service zone name: Type (Hospital/EmergencyShelter/Residential/EmergencyService): Requested water (L): Minimum requirement (L): Distribution loss (%):
Previous waiting cycles:
--- Zones ordered by allocation priority ---
Rank Service Zone Reqstd MinReq Alloc EffDlv Shortage Condition
-----
1 Municipal Hospital 4000.0 3000.0 3477.1 3407.6 592.4 MinimumSatisfied
2 Ward 5 Residential Area 3800.0 1800.0 2045.5 1800.0 2000.0 MinimumSatisfied
3 Central Flood Shelter 3500.0 2500.0 2604.2 2500.0 1000.0 MinimumSatisfied
4 School Emergency Shelter 2600.0 1600.0 1684.2 1600.0 1000.0 MinimumSatisfied
5 Fire and Emergency Service 2000.0 1500.0 1515.2 1500.0 500.0 MinimumSatisfied
6 Ward 3 Residential Area 4500.0 2000.0 2173.9 2000.0 2500.0 MinimumSatisfied

===== SUMMARY OF ALLOCATION =====
Zone ID | Service Zone | Requested | MinReq | Priority | Allocated | EffDelivered | Shortage | Condition
-----
Z01 | Municipal Hospital | 4000.0 | 3000.0 | 1 | 3477.1 | 3407.6 | 592.4 | MinimumSatisfied
Z04 | Ward 5 Residential Area | 3800.0 | 1800.0 | 2 | 2045.5 | 1800.0 | 2000.0 | MinimumSatisfied
Z02 | Central Flood Shelter | 3500.0 | 2500.0 | 3 | 2604.2 | 2500.0 | 1000.0 | MinimumSatisfied
Z05 | School Emergency Shelter | 2600.0 | 1600.0 | 4 | 1684.2 | 1600.0 | 1000.0 | MinimumSatisfied
Z06 | Fire and Emergency Service | 2000.0 | 1500.0 | 5 | 1515.2 | 1500.0 | 500.0 | MinimumSatisfied
Z03 | Ward 3 Residential Area | 4500.0 | 2000.0 | 6 | 2173.9 | 2000.0 | 2500.0 | MinimumSatisfied

Total water available at start : 13500.0 L
Total water requested : 20400.0 L
Total water allocated (released) : 13500.0 L
Total effective water delivered : 12807.6 L
Total unresolved requirement : 7592.4 L
Remaining water at distribution centre: 0.0 L
Zones fully satisfied : 0
Zones with only minimum satisfied : 6
Zones below minimum requirement : 0
Zones completely unserved : 0
```